Tutorial Sheet 7

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Instructions

This sheet is designed to invoke thinking and careful consideration of the mathematics you have learnt. It is not designed for speed. It is designed for depth.

You are expected to:

- justify every algebraic and probabilistic step,
- explain what you are trying to compute before computing it,
- define every statistical term you use,
- interpret answers in context.

Important

The following are **not** accepted:

- "calculator says",
- skipping distribution assumptions,
- giving a p-value without conclusion.

Being stuck is acceptable. Lack of attempt is not.

1 Warm Up

1. Define clearly:

   a) mean,

   b) median,

   c) mode,

   d) variance,

   e) standard deviation,

   f) expectation value,

   g) interquartile range.

2. Explain the difference between:

   a) discrete and continuous random variables,

   b) population and sample,

   c) parameter and statistic.

3. Suppose a system S has N constituents where it is distributed probabilistically. If each constituents has probability of p_i, what is the value of $\sum_i p_i$?

4. A dataset has mean 12 and standard deviation 3. All values are increased by 5.

   a) What happens to the mean?

   b) What happens to the standard deviation?

   c) Explain why.

5. Suppose two events, A and B - Explain in words what $A \cup B$ represents. Hence or otherwise, state the addition laws:

   $$P(A \cup B) = ?$$

   in both the general case and when events are mutually exclusive.

6. Explain what it means for two events to be independent.

7. Scientists at CERN, Switzerland [1] claims are up to 5σ significance. What does this mean and why do they have this convention?

[1] CERN is a world-wide collaboration working on High Energy Particle Physics, including particle accelerators. For context, see: this linked page)

2 Probability Foundations

Theory

For events A and B:

$$P(A \cap B) = P(A)P(B|A)$$

If independent:

$$P(A \cap B) = P(A)P(B)$$

Conditional probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

provided $P(B) > 0$.

1. A box contains 4 red and 6 blue balls. Two are drawn without replacement.

   a) Find $P(\text{both red})$.

   b) Are the events independent? Justify.

2. A disease test is 98% accurate for infected individuals and 95% accurate for non-infected individuals. 1% of the population has the disease.

   a) Construct a probability tree.

   b) Find $P(\text{disease} \mid \text{positive})$.

   c) Comment on the result.

3. Prove from first principles that if A and B are independent, then A and B^c are independent.

4. Three events A, B, C are pairwise independent. Does this imply mutual independence? Provide either proof or counterexample.

3 Binomial Distribution

Theory

A random variable $X \sim \mathrm{B}(n, p)$ if:

- fixed number n of trials,
- independent trials,
- two outcomes,
- constant probability p.

Probability of $X = r$ is given by:

$$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$$

Expectation value of X is given by:

$$\mathbb{E}(X) = np, \quad \mathrm{Var}(X) = np(1-p)$$

1. Rewrite the formula $P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$ into ${}^nC_r$ and explain why ${}^nC_r$ naturally appears within this formula - what is common about both?

2. A biased coin has probability 0.3 of heads. It is tossed 12 times.

   a) Find $P(X = 4)$.

   b) Find $P(X \geq 8)$.

   c) Find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

3. A factory claims defect rate is 5%. In a sample of 40 items, 6 are defective.

   a) State an appropriate binomial model.

   b) Find $P(X \geq 6)$ under the claim.

c) Comment informally on whether 6 seems unusually large.

4. Show algebraically that:

$$\sum_{r=0}^{n} \binom{n}{r} p^r (1-p)^{n-r} = 1.$$

5. A binomial random variable has mean 6 and variance 3.6. Find n and p.

4 Normal Distribution

Theory

If $X \sim N(\mu, \sigma^2)$ then:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1).$$

1. Recall the formula $Z = \frac{X-\mu}{\sigma}$, explain what each symbol represents. Hence or otherwise, draw rough sketch to demonstrate the shape of this distribution and comment on how it compares to Binomial Distribution.

2. Heights are normally distributed. In the UK, it has mean of 175 cm and 162 cm respectively for biological male and female, with standard deviation around 8 cm.

   a) Find $P(X > 185)$ for a biological female.

   b) Find the 90th percentile for both biological male and female

   c) State a height of your own choice and hence, find its corresponding percentile.

3. A machine fills bottles with mean 500 ml and sd 4 ml.

   a) Find probability a bottle contains less than 492 ml.

   b) If 100 bottles are sampled, approximate distribution of sample mean.

4. Explain why continuity correction is needed when approximating a binomial distribution using a normal distribution.

5. A random variable is approximately $N(10, 4)$. Find k such that:

$$P(10 - k < X < 10 + k) = 0.95.$$

5 Hypothesis Testing

Theory

Steps:

1. State H_0 and H_1.
2. Choose significance level α.
3. Identify distribution under H_0.
4. Compute test statistic.
5. Compare with critical region or compute p-value.
6. Conclude in context.

1. A coin is suspected to be biased towards heads. It is tossed 20 times and gives 15 heads.

   a) State hypotheses.

   b) Perform test at 5% level.

   c) State conclusion clearly.

2. A manufacturer claims mean lifetime is 1200 hours. A sample of 36 bulbs has mean 1170 hours. Population sd known to be 90 hours.

   a) State hypotheses (two-tailed).

   b) Calculate test statistic.

   c) Conclude at 1% level.

3. Explain Type I and Type II errors in words.

4. A binomial test is carried out at 5% level. Explain what 5% means in practical terms.

5. In a one-tailed test, the p-value is 0.032. Interpret this fully in context.